Instrumented & Monitored Cloud Service

Observability Monitoring

Cloud Engineer / Ironhack

Faramarz Karamizadeh, Aug.2026

Agenda

1. 1. Architecture & Instrumentation
2. 2. Live Demo — Dashboard & Monitoring
3. 3. Incident Response Simulation
4. 4. Alerting & Response
5. 5. Learnings & Improvements
6. 6. Q&A

16 widgets

• Period 60s • SingleValue + Golden Signals + Correlation

5 Custom metrics

request_count • api_latency • error_count • orders_per_minute • order_value

Architecture

INSTANCE

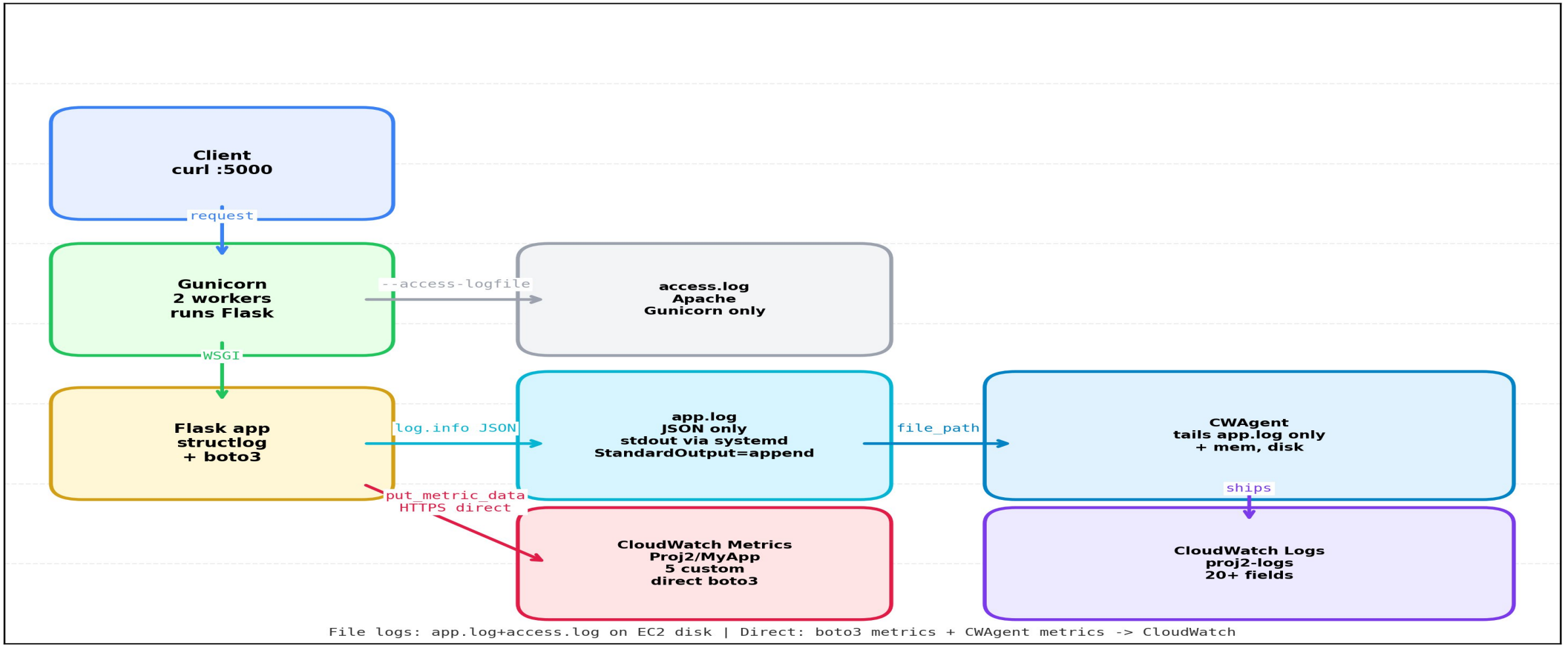
i-01bxxx myapp.service • Flask + Gunicorn 2 sync workers. Systemd service

METRICS

Namespaces: **Proj2/MyApp** (custom) + CWAgent (mem/disk) + AWS/EC2 (CPU/Network) • boto3 put_metric_data • period 60s

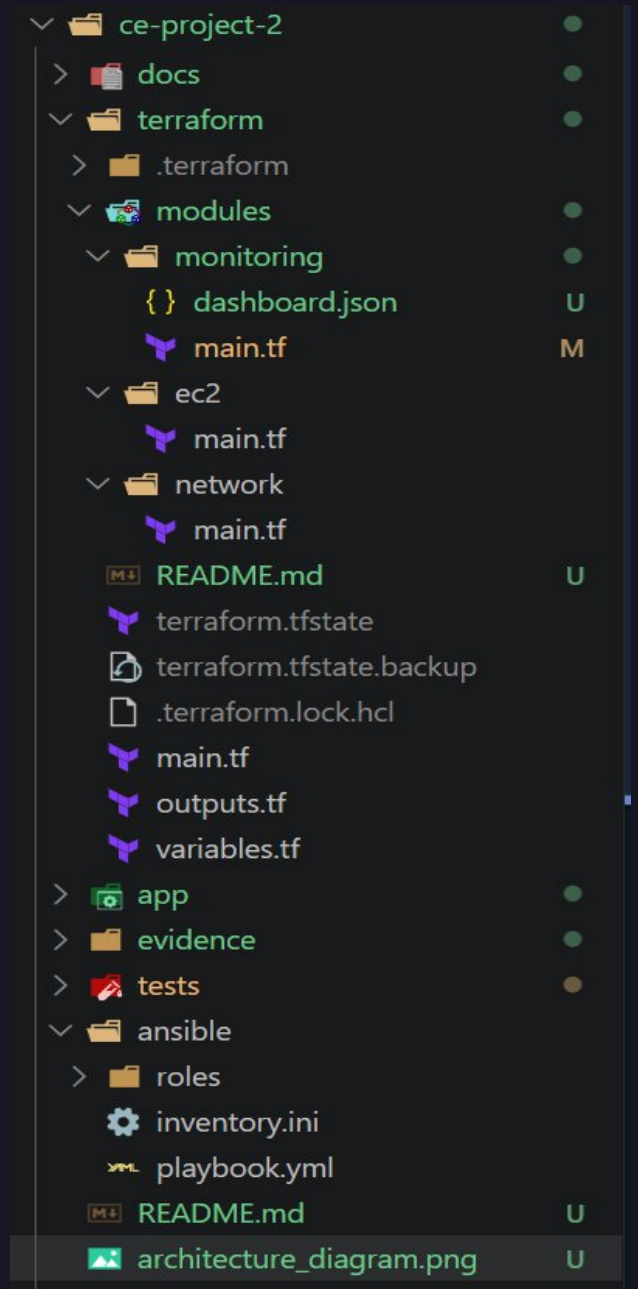
LOGGING PIPELINE

systemd StandardOutput → **app.log JSON only** • gunicorn-error.log • CWAgent → **proj2-logs** log group



Project Highlights

100% IaC • Terraform + Ansible



Infra as Code 100%

Terraform modular + Ansible full stack

VPC, EC2 i-01b04296, SG, IAM-role, CWAgent configs, myapp.service. Reproducible.

5 Custom Metrics [Business]

orders_per_minute + order_value

request_count api_latency error_count + biz orders_per_minute order_value via boto3.

6 Warn / Critical Alarms

high-error, high-latency, no-traffic, high-cpu

>5 errors /2x60s, P95 >800ms, no traffic 5m, cpu >85% → SNS proj2-alerts.

Structured Logging

correlation_id • level • latency_ms

structlog JSON → 20+ fields. X-Corr-ID 51e9ef29 → filter correlation_id='demo-123'.

Custom Widgets

SingleValue + USE

Rate, Duration, Errors, Saturation unified.

Memory Anomaly alarm and dashboard [USE]

mem_used_percent detection

Infra saturation signal from CWAgent. SingleValue widget + anomaly band vs latency.

Golden signals

Traffic→ Error→ Latency→ Saturation

EC2 → Gunicorn → Flask → CWAgent → CloudWatch - full traceability

Composite alarms

combining multiple conditions

mem-use-anomaly-detection & high-latency

Correlation Dashboard

Latency vs Traffic vs Errors vs CPU

One view: P95 latency ↔ request_count ↔ error_count ↔ CPU. Golden + USE.

Custom Metrics — 5 signals via boto3 put_metric_data

request_count

COUNT • Sum • after_request

Every request increments. Fuels Rate & health. Never rely on ALB logs.

api_latency

MILLISECONDS • Avg/Max/P95

latency_ms = (now - g.start)*1000. Duration golden signal.

error_count

COUNT • Sum • if status >=400

Errors golden signal. Triggers high-error alarm when >5 in 2x60s.

orders_per_minute

Business saturation signal. Sum per minute.

BUSINESS

order_value

Amount metric. Tracks revenue proxy, no unit.

None

Alerting — 6 Alarms CloudWatch

error-rate

error_count • Sum >5 • 2 periods of 60s

RED Errors | Stat Sum | Treat missing as not-breaching

latency

api_latency • Average >500ms • 2 periods

RED Duration | Avg | SLO 500ms

memory

request_count • Sum <1 • 300s period • missing data BREACHING

Health-check | 1 req /5min min | Missing → ALARM

disk

DISKUtilization • Average >80% • 2 periods 300s • Dim: InstanceId=i-01b042…

USE Saturation | AWS/EC2 | Avg 80%

memory anomaly

request_count • Sum <1 • 300s period • missing data BREACHING

Health-check | 1 req /5min min | Missing → ALARM

Composite

anomaly + error rate

Health-check | 1 req /5min min | Missing → ALARM

- **SNS → proj2-alerts + Email notification** • IAM proj2-ec2-role needs cloudwatch:PutMetricData only; ListMetrics from Admin • Thresholds tuned for low-traffic demo, not prod scale

Structured Logging

OUTPUT — app.log / journalctl -u myapp


```
{
"event": "order_created",
"order_id": "09bcdd18-fd61-48b5-94a8...",
"amount": 150,
"correlation_id": "51e9ef29-4528-4582-a712...",
"timestamp": "2026-08-13T12:58:26.558Z"
}
{
"event": "request_completed",
"correlation_id": "51e9ef29...",
"method": "POST",
"path": "/orders",
"status_code": 201,
"latency_ms": 41,
"level": "INFO",
"timestamp": "2026-08-13T12:58:26.599Z"
}
```

Live Demo Plan — proj2-dashboard

DASHBOARD

proj2-dashboard • 12 widgets • Period 60s • Region us-east-1 • Auto-refresh 60s

QUICK STATS

SingleValue → timeSeries → correlation. All three namespaces: Proj2/MyApp + CWAgent + AWS/EC2

01 — TOP ROW

SingleValue at a glance

Current rate for decision in 3 seconds

- request_count Sum
- error_count Sum
- api_latency Avg
- orders/min Sum

02 — MIDDLE

Golden Signals

RED mapping: traffic, errors, duration, saturation

- Traffic request_count
- Errors error_count
- Latency Avg/Max
- CPU + Orders

03 — BOTTOM

EC2 USE Resources

Utilization + Saturation. Disk uses SEARCH

- CPUUtilization
- mem_used_percent
- Network In/Out
- SEARCH disk% → [[ ]]

04 — CORRELATION

Latency vs Traffic vs Errors vs CPU

Overlay all in one graph. When /fail injected: Error spikes, Latency flat, CPU flat → app error not infra

- api_latency
- request_count
- error_count
- CPU %

Instrumentation — Key Decitions

STRUCTURED LOGGING

structlog JSONRenderer + TimeStamper ISO

```
log.info("request_completed", correlation_id, method, path, status_code, latency_ms, level)
```

Decision: CloudWatch Logs Insights auto-parses JSON → 20+ fields

CORRELATION

before_request → correlation_id

```
g.correlation_id = request.headers.get("X-Correlation-ID", uuid4())
g.start_time = time()
```

Every log line carries same ID → filter correlation_id="demo-123" finds full flow.

CUSTOM METRICS

boto3 put_metric_data • Namespace Proj2/MyApp

```
cloudwatch.put_metric_data(
 Namespace="Proj2/MyApp",
 MetricData=[{Name: "request_count", Value:1, Timestamp:utcnow()}])
```

5 signals in after_request + business hook.

ERROR LEVEL

3 levels of error

WARN-ERROR-INFO

RED / USE Analysis — how we triaged

DASHBOARD MAPPING

RED — SERVICE VIEW

Rate	request_count Sum • Traffic timeSeries
Errors	error_count Sum • red spike
Duration	api_latency Avg / Max / P95

USE — RESOURCE VIEW

Utilization	CPUUtilization + mem_used% + disk SEARCH [[ ]]
Saturation	orders_per_minute • business queue
Errors	same error_count • overlay correlation

Evidence: Injected /fail 20x → Error Rate ↑ 5+ in 60s, Latency flat 30ms, CPU 8% flat → **app error not infra**. Top row = RED single value, Middle = Golden Signals, Bottom = USE, Correlation = all overlayed.

Learnings — What I Learned

01

Separation of concerns: access.log vs app.log

Pointing both **--access-logfile** and **--error-logfile** to same **app.log** mixes Apache format 91.0.58.40 - - [13/Aug...] with JSON. Parser breaks → Discovered fields 11, no level.

02

IAM least privilege bite

proj2-ec2-role PutMetricData works but ListMetrics fails. Validate from Admin console, not EC2. Learned to test both planes.

03

Correlation ID is game changer

One header **X-Correlation-ID** = **51e9ef29-4528…** ties order_created + request_completed. Without X-Ray, full trace in Logs Insights via **filter correlation_id="demo-123"**.

What Would I Do Differently?

01 — INSTRUMENTATION

Use OpenTelemetry SDK, not manual boto3 put_metric_data

Dimensions out of the box, context propagation, no try/except hand-rolled. OTEL → CloudWatch exporter keeps native but adds trace id.

02 — USING GRAFANA

Using a third-parthy opensource dashboard

It cloud be a good practice for multi-cloud environmet with no dependency on Cloudwatch

Thank You

Q&A